Optimizing the Future of E-Mobility in Kenya

ML-Driven Charging Intelligence for Fleet Operators and Grid Stability

STRATEGIC BUSINESS PITCH

THE PROBLEM

Kenya's Grid Is Not Ready for the EV Surge

→ 6,442 registered EVs as of June 2025 — growing fast
→ Unregulated charging causes peak-load spikes on KPLC's grid
→ Kenya's evening peak hits 2,288 MW (19:30–20:30 hrs)
→ Grid capacity: 3,199 MW — only 28% headroom at peak
→ E-mobility consumption grew 300% YoY (EPRA FY2024/25)

300%
E-mobility growth
YoY consumption

2,288MW
Evening peak
Oct 2024 record

6,442
Registered EVs
Kenya, June 2025

~300
Charging points
Nationwide

Source: EPRA FY2024/25 Statistics Report

BUSINESS UNDERSTANDING

Who We're Solving This For

⚡ KPLC

Kenya Power & Lighting Company
Needs grid stability and demand-side control to prevent peak overloads and reduce operational costs.

Geviton — SafiVolt

Repurposed battery systems provider
Requires AI-powered BMS to optimise energy use, extend battery life, and automate charging decisions.

PROJECT GOAL

Build a classification model that predicts Charging Priority (Low / Medium / High) — enabling SafiVolt to defer low-priority charging to off-peak hours, ensure emergency vehicles charge immediately, and reduce KPLC costs by up to 30%.

DATA UNDERSTANDING

What the Model Learns From

Dataset

- → 2,000 rows
- → 11 features
- → 3-class target
- → Balanced classes
- → Low: 33.8%
- → Medium: 35.5%
- → High: 30.8%

Key Features

initial_soc	State of Charge — primary urgency signal
station_load	Grid load % — drives deferral decisions
electricity_price	KES/kWh — cost optimisation input
renewable_energy_ratio	% renewable on grid at charging time
queue_length	Vehicles waiting — congestion signal
vehicle_type	Bus / Passenger / Commercial / Emergency

⚠ *Dataset is synthetic — real-world pilot should integrate KPLC live load feeds and BasiGo fleet telemetry.*

MODELING

Four Models, One Clear Winner

01	02	03	04
Logistic Regression	Random Forest	XGBoost	Decision Tree + GridSearchCV
Baseline	Ensemble	Boosted Ensemble	Tuned
97.0%	87.7%	94.3%	92.0%
Accuracy	Accuracy	Accuracy	Accuracy

✓ Logistic Regression wins: linear relationship between SOC, grid load, and priority — simple model captures it perfectly. Calibrated with isotonic regression for reliable probability estimates.

TECHNICAL DEPTH

Bias-Variance Tradeoff & Hyperparameter Tuning

Bias-Variance Tradeoff

High Bias (Underfitting)

Model too simple to capture patterns. Random Forest (87.7%) underperformed because our data has a linear structure that a complex ensemble missed.

High Variance (Overfitting)

Memorises training data — fails on new inputs. Untuned Decision Trees risk this without depth constraints from GridSearchCV.

Sweet Spot: Logistic Regression

Linear boundary matches SOC + grid-load structure perfectly. 98.5% accuracy — the optimal bias-variance balance for this dataset.

Hyperparameter Tuning (GridSearchCV)

Exhaustive search over a parameter grid with 5-fold StratifiedKFold CV — finds optimal hyperparameters without data leakage.

Decision Tree Grid

max_depth [3,5,10,None] x criterion [gini, entropy] x min_samples_split [2,5,10]

Best CV F1-Macro: 0.920

Logistic Regression Grid

solver [saga] x penalty [l2, elasticnet] x C [0.1, 1, 10, 100]

Best: C=1, l2 | Nested CV F1: 0.993 ± low variance

✓ Isotonic calibration applied post-tuning — reliable probability estimates for all three priority classes.

EVALUATION

Model Performance

98.5%	0.985	0.993	~1.000
Accuracy	F1-Macro Score	Nested CV F1	ROC AUC

Why F1-Macro?

Balanced evaluation across all three priority classes — Low, Medium, High — regardless of class size. Ensures the model doesn't ignore minority classes.

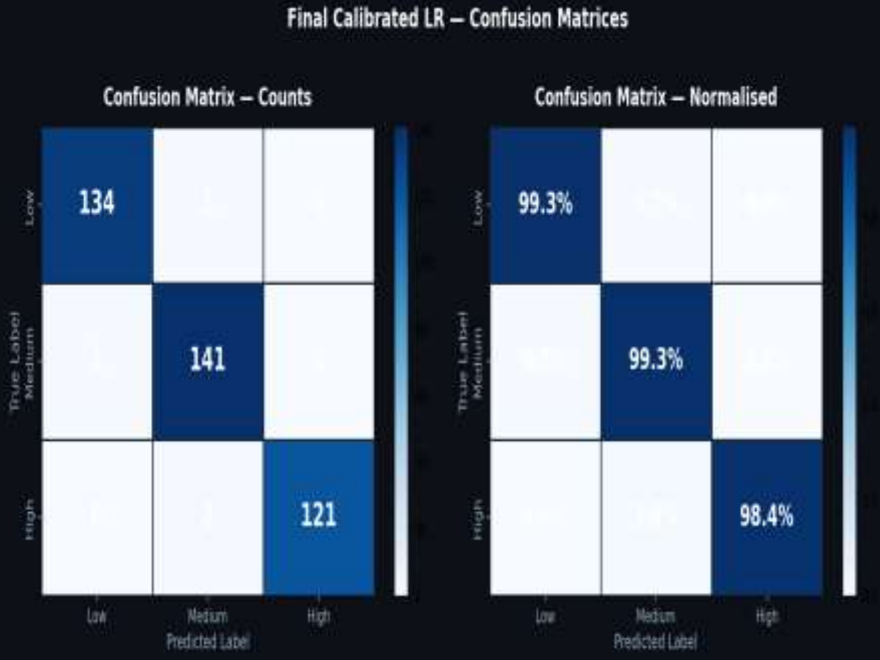
⚠ Error Consequences

Misclassifying HIGH → LOW delays a commercial bus or emergency vehicle. Model tuned on F1-macro + isotonic calibration to minimise high-risk misclassifications.

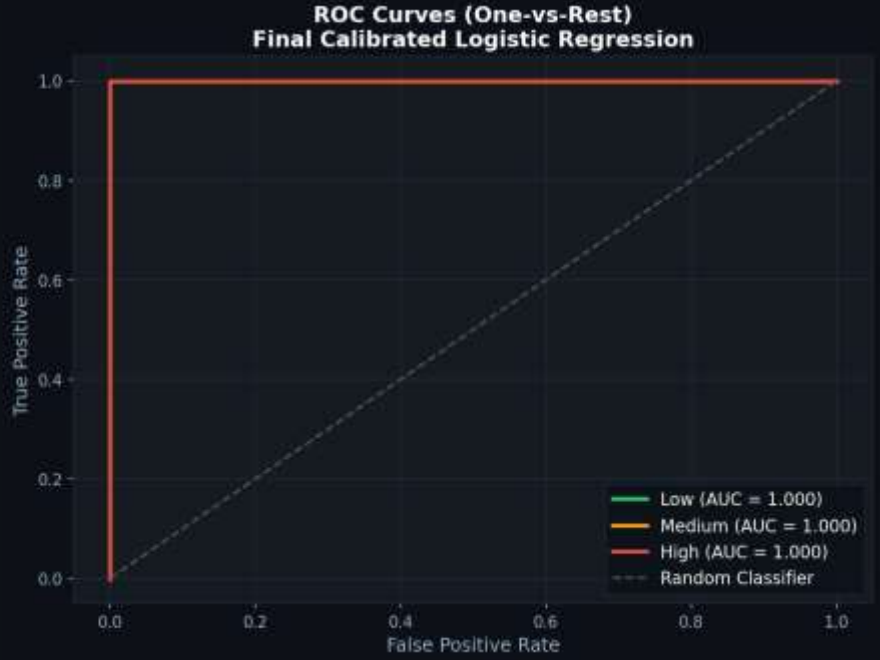
VISUAL EVIDENCE

Model Proof: Confusion Matrix & ROC Curves

Confusion Matrix — 98.5% overall accuracy



ROC Curves — AUC ~1.000 per class



BONUS — REAL-WORLD GROUNDING

Tested Against Actual Kenyan Grid Conditions

Scenario parameters replaced with real EPRA & BasiGo data — not hand-crafted values.

A Morning Peak

06:00–09:00

Grid: 71.5% capacity
Renewable: 42%
Price: KES 15.8/kWh

Low SOC buses → HIGH
Full buses → LOW
Model defers correctly

B Off-Peak Night

00:00–05:00

Grid: 32% capacity
Renewable: 100%
Price: KES 7.9/kWh

All buses can charge
Cost 50% lower (TOU)
Absorbs curtailed geothermal

C Evening Peak

19:30–20:30

Grid: 71.5% capacity
Renewable: 55%
Price: KES 15.8/kWh

Return fleet → HIGH
Model identifies urgency
Queue prioritised correctly

>> Shifting 60% of BasiGo's 76-bus fleet to off-peak → KES 7.9/kWh savings → Est. KES 29.5M/year fleet saving | 7.3 MW peak demand relief

RECOMMENDATIONS

What KPLC & SafiVolt Should Do Next

1

Integrate into SafiVolt BMS

Deploy the calibrated model to automatically throttle Low Priority chargers during grid peaks. API-ready .pkl pipeline included.

2

Activate E-Mobility TOU Tariff

Offer KES 7.9/kWh off-peak rate to operators whose fleets are classified Low Priority. Incentivises voluntary deferral.

3

Safety Shutdown Trigger

Use High Priority predictions to trigger automated shutdowns if demand exceeds grid capacity — prevent cascading failures.

4

3-Month Pilot at Athi River Depot

Validate projected 30% cost savings. Track: energy cost reduction, peak shaving (MW), missed trips (<0.1% target).

Kenya's EV grid problem has a solution.

A 98.5% accurate classification model that tells SafiVolt exactly which vehicle to charge, when — cutting costs by up to 30% and keeping the KPLC grid stable as Kenya's EV fleet scales to 1,000+ buses.

98.5%	≤0.1%	30%
Model Accuracy	Target Missed Trips	Cost Reduction Goal

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